

Problem 1

$$Y_i = \beta_0 + \gamma W_i + \varepsilon_i$$

a) W.K.T, $x'y = (x_1 \dots x_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \sum_{i=1}^n x_i y_i$

$$1'1 = (1 \ 1 \ 1 \dots 1) \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = 1+1+\dots (n \text{ times}) \\ = \sum_{i=1}^n 1 = n$$

$$1'W = (1 \ 1 \dots 1) \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \\ = \sum_{i=1}^n w_i = n_1 \text{ (no. of treated units)}$$

$$W'W = (w_1 \dots w_n) \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \\ = \sum_{i=1}^n w_i^2 = \sum_{i=1}^n w_i \quad \left(\text{since } w_i = \begin{cases} 1, & \text{treated} \\ 0, & \text{control} \end{cases} \right) \\ = n_1$$

b) Given, $X = \begin{pmatrix} 1 & w_1 \\ \vdots & \vdots \\ 1 & w_n \end{pmatrix}$

$$X'X = \begin{pmatrix} 1 & 1 & \dots & 1 \\ w_1 & w_2 & \dots & w_n \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \\ w_1 \\ \vdots \\ w_n \end{pmatrix} \quad (\text{from part a}) \\ = \begin{pmatrix} \sum_{i=1}^n 1 & \sum_{i=1}^n w_i \\ \sum_{i=1}^n w_i & \sum_{i=1}^n w_i^2 \end{pmatrix} = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}$$

c) Given, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

then, $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

from part b, we have

$$X'X = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}$$

So, $a = n$
 $b = c = d = n_1$

$$(X'X)^{-1} = \frac{1}{n \cdot n_1 - n_1 \cdot n_1} \begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix}$$

$$= \frac{1}{n_1(n_0 - n_1)} \begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix}$$

$$= \frac{1}{n_1 n_0} \begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{n_0} & -\frac{1}{n_0} \\ -\frac{1}{n_0} & \frac{n}{n_1 n_0} \end{pmatrix}$$

d) Given, $y' = (y_1 \dots y_n)$ where

$$y_i = \beta_0 + \gamma w_i + \epsilon_i$$

$$\cancel{X'y} = (\cancel{x_1} \dots \cancel{x_n}) \begin{pmatrix} \cancel{y_1} \\ \vdots \\ \cancel{y_n} \end{pmatrix}$$

$$\begin{aligned}
 X'y &= \begin{pmatrix} 1 & 1 & \dots & 1 \\ w_1 & w_2 & \dots & w_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\
 &= \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n w_i y_i \end{pmatrix}
 \end{aligned}$$

e) from part d, we have

$$X'y = \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n w_i y_i \end{pmatrix}$$

$$\begin{aligned}
 \sum_{i=1}^n y_i &= \sum_{(w_i=0)} y_i + \sum_{(w_i=1)} y_i \\
 &= n_0 \bar{y}^{obs}(0) + n_1 \bar{y}^{obs}(1)
 \end{aligned}$$

$$\begin{aligned}
 \sum_{i=1}^n w_i y_i &= \sum_{(w_i=0)} w_i y_i^0 + \sum_{(w_i=1)} w_i y_i \\
 &= \sum_{(w_i=1)} y_i = n_1 \bar{y}^{obs}(1)
 \end{aligned}$$

$$\text{So, } X'y = \begin{pmatrix} n_0 \bar{y}^{obs}(0) + n_1 \bar{y}^{obs}(1) \\ n_1 \bar{y}^{obs}(1) \end{pmatrix}$$

f) Need to show that,

$$\hat{\gamma}_{unadj} = \bar{y}^{obs}(1) - \bar{y}^{obs}(0)$$

least square estimators (slide 13, Set 3),

$$\hat{\beta} = (X'X)^{-1} X'y$$

from part c, $(X'X)^{-1} = \begin{pmatrix} \frac{1}{n_0} & \frac{-1}{n_0} \\ \frac{-1}{n_0} & \frac{n}{n_1 n_0} \end{pmatrix}$

from part d, $X'y = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n W_i Y_i \end{pmatrix}$

$$= \begin{pmatrix} n_0 \bar{Y}^{obs}(0) + n_1 \bar{Y}^{obs}(1) \\ n_1 \bar{Y}^{obs}(1) \end{pmatrix}$$

$$\hat{\beta} = \begin{pmatrix} 1/n_0 & -1/n_0 \\ -1/n_0 & n/(n_1 n_0) \end{pmatrix}_{2 \times 2} \begin{pmatrix} n_0 \bar{Y}^{obs}(0) + n_1 \bar{Y}^{obs}(1) \\ n_1 \bar{Y}^{obs}(1) \end{pmatrix}_{2 \times 1}$$

$$= \frac{1}{n_0} \left(n_0 \bar{Y}^{obs}(0) + n_1 \bar{Y}^{obs}(1) - \frac{1}{n_0} (n_0 \bar{Y}^{obs}(0) + n_1 \bar{Y}^{obs}(1)) \right)$$

$$+ \left(\frac{-1}{n_0} \right) \left(n_0 \bar{Y}^{obs}(0) + n_1 \bar{Y}^{obs}(1) \right)$$

$$+ \left(\frac{n}{n_1 n_0} \right) \left(n_1 \bar{Y}^{obs}(1) \right)$$

$$\hat{\tau}_{unadj} = -\bar{Y}^{obs}(0) - \frac{n_1}{n_0} \bar{Y}^{obs}(1)$$

$$= -\bar{Y}^{obs}(0) + \bar{Y}^{obs}(1) \left(\frac{n}{n_0} - \frac{n_1}{n_0} \right)$$

$$= \bar{y}_{obs}(1) - \bar{y}_{obs}(0)$$

HW9

2025-04-18

Problem 2

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
#was not able to download directly, did save as
data <- read.csv("/Users/manideeppendyala/Downloads/D100F07.csv")
head(data)
```

```
##   v_pres_general_04 v_cong_general_06 v_pres_general_08 age evervoted regyear
## 1                1                0                0 29          1    2008
## 2                1                1                1 54          1    1987
## 3                1                0                1 71          1    1960
## 4                0                0                1 52          1     NA
## 5                1                1                1 53          1    1979
## 6                0                1                1 23          1    2005
##   s_r_v badaddress town1_block town2_block town3_block town4_block town5_block
## 1     3        NA           0           0           0           0           0
## 2     3        NA           0           0           0           0           0
## 3     3        NA           0           0           0           0           0
## 4     3        NA           0           0           0           0           0
## 5     3        NA           0           0           0           0           1
## 6     3        NA           0           0           0           0           0
##   town6_block treatment age_sq sampled_strata_1 sampled_strata_2
## 1           0        NA    841                0                0
## 2           0        NA   2916                0                0
## 3           1        NA   5041                0                0
## 4           0        NA   2704                0                0
## 5           0        NA   2809                0                0
## 6           0        NA   529                 0                0
##   sampled_strata_3 i_grp_addr_1 i_grp_addr_2 i_grp_addr_3 i_grp_addr_4 dem rep
## 1                1            0            0            1            0    0    0
```

```
## 2      1      0      1      0      0 0 1
## 3      1      0      0      1      0 0 1
## 4      1      1      0      0      0 0 1
## 5      1      0      0      0      1 1 0
## 6      1      1      0      0      0 1 0
##   sex_1 sex_2 sex_3 badaddress_v2 returnedmail foundvoterecord voted_10g
## 1     1     0     0           0          NA           1           1
## 2     0     1     0           0          NA           1           1
## 3     0     1     0           0          NA           1           1
## 4     0     0     1           0          NA           1           0
## 5     1     0     0           0          NA           1           1
## 6     1     0     0           0          NA           1           1
```

#Part a

```
data_clean <- data %>%
  filter(badaddress == 0)

data_clean <- data_clean %>%
  mutate(
    treated = ifelse(treatment %in% c(5, 6, 7), 1, 0),
    block = s_r_v,
    outcome = voted_10g
  )

blocks <- data_clean %>%
  group_by(block) %>%
  summarise(
    n_b = n(),
    n1 = sum(treated == 1),
    n0 = sum(treated == 0),
    Y1_bar = mean(outcome[treated == 1]),
    Y0_bar = mean(outcome[treated == 0]),
    s1_sq = var(outcome[treated == 1]),
    s0_sq = var(outcome[treated == 0]),
    tau_b = Y1_bar - Y0_bar,
    var_tau_b_ney = (s1_sq / n1) + (s0_sq / n0)
  ) %>%
  select(block, n_b, tau_b, var_tau_b_ney)

print(blocks)
```

```
## # A tibble: 3 x 4
##   block  n_b  tau_b var_tau_b_ney
##   <int> <int> <dbl>      <dbl>
## 1     1 69488 0.0341    0.0000684
## 2     2 68859 0.0203    0.000353
## 3     3 18586 0.00255    0.0000660
```

#Part b


```

N_total <- sum(blocks$n_b)
tau_hat <- sum(blocks$n_b * blocks$tau_b) / N_total
var_tau_hat <- sum((blocks$n_b^2 / N_total^2) * blocks$var_tau_b_nei)

cat("\nEstimated Average Treatment Effect:", round(tau_hat, 4), "\n")

##
## Estimated Average Treatment Effect: 0.0243

cat("Estimated Variance of Estimated Average Treatment Effect:", round(var_tau_hat, 6), "\n")

## Estimated Variance of Estimated Average Treatment Effect: 8.2e-05

#part c

z <- tau_hat / sqrt(var_tau_hat)
p_val <- 1 - pnorm(z)

cat("\nZ-statistic:", round(z, 4), "\n")

##
## Z-statistic: 2.6782

cat("One-sided p-value:", round(p_val, 4), "\n")

## One-sided p-value: 0.0037

if (p_val < 0.01) {
  cat("We reject Null hypothesis as p-value is less than 0.01\n")
} else {
  cat("We fail to reject Null hypothesis\n")
}

## We reject Null hypothesis as p-value is less than 0.01

ci_lower <- tau_hat - 1.96 * sqrt(var_tau_hat)
ci_upper <- tau_hat + 1.96 * sqrt(var_tau_hat)

cat("\n95% Confidence Interval for Tau: (", round(ci_lower, 4), ",", round(ci_upper, 4), ")\n")

##
## 95% Confidence Interval for Tau: ( 0.0065 , 0.0421 )

Problem 3

library(speff2trial)

## Loading required package: leaps

## Loading required package: survival

```



```
library(ggplot2)
library(cobalt)
```

```
## cobalt (Version 4.6.0, Build Date: 2025-04-15)
```

```
library(sandwich)
```

```
data(ACTG175)
df <- ACTG175 %>%
  select(cd420, treat, age, wtkg, hemo, gender)
df$treat <- factor(df$treat, levels = c(0, 1), labels = c("Monotherapy", "Combination"))
head(df)
```

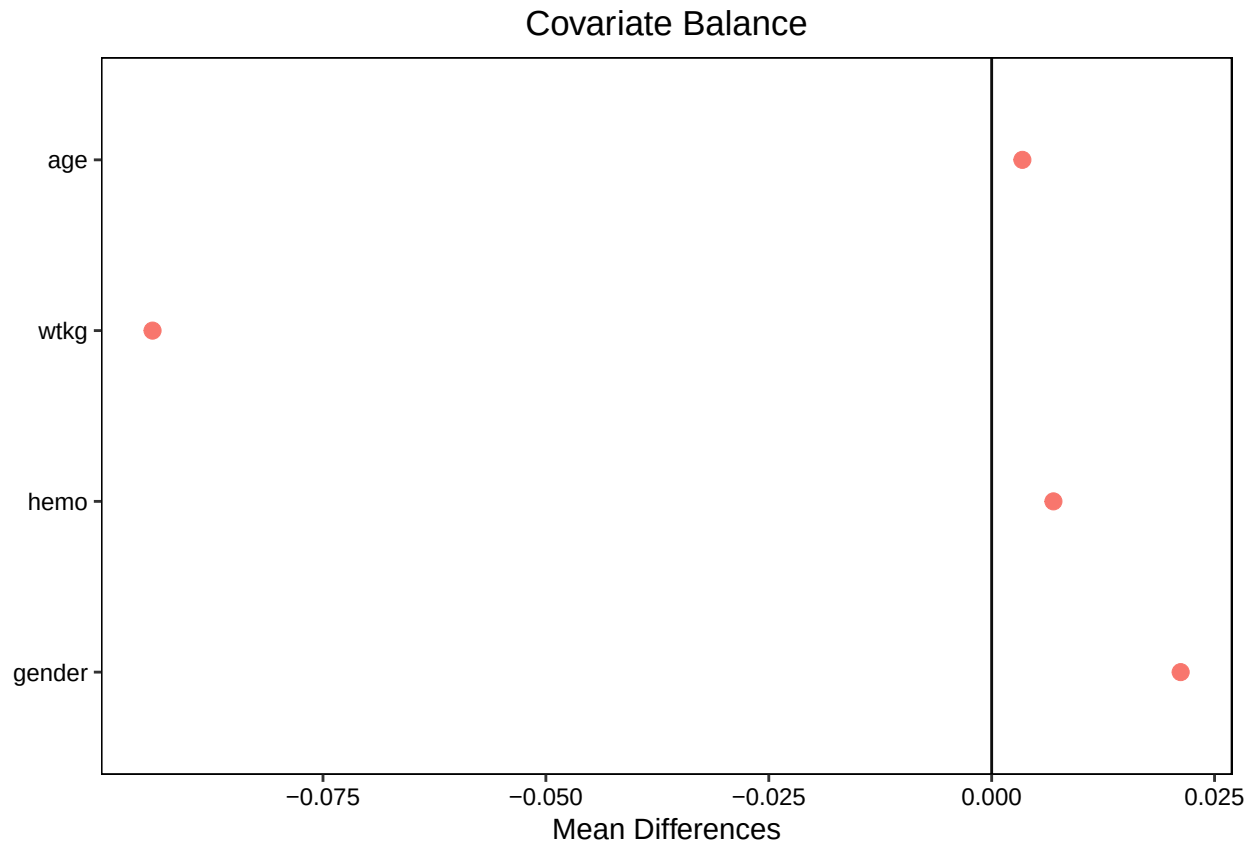
```
##   cd420      treat age   wtkg hemo gender
## 1   477 Combination 48 89.8128   0      0
## 2   218 Combination 61 49.4424   0      0
## 3   274 Combination 45 88.4520   0      1
## 4   394 Combination 47 85.2768   0      1
## 5   353 Monotherapy 43 66.6792   0      1
## 6   339 Combination 46 88.9056   0      1
```

```
#part a
```

```
love.plot(
  x = subset(df, select = c(age, wtkg, hemo, gender)),
  treat = df$treat
) +
theme(legend.position = "none")
```

```
## Note: 's.d.denom' not specified; assuming "pooled".
```

```
## Warning: Standardized mean differences and raw mean differences are present in
## the same plot. Use the 'stars' argument to distinguish between them and
## appropriately label the x-axis. See '?love.plot' for details.
```



#part b

```
unadjmodel <- lm(cd420 ~ treat, data = df)
tau_hat <- coef(unadjmodel)[2]

mono <- df$cd420[df$treat == "Monotherapy"]
combo <- df$cd420[df$treat == "Combination"]

n0 <- length(mono)
n1 <- length(combo)
var0 <- var(mono)
var1 <- var(combo)
var_tau_hat <- var0/n0 + var1/n1
se_tau_hat <- sqrt(var_tau_hat)

cat("Tau hat:", tau_hat, "\n")
```

```
## Tau hat: 46.8105
```

```
cat("Variance:", var_tau_hat, "\n")
```

```
## Variance: 45.70026
```

```
cat("Standard Error:", se_tau_hat, "\n\n")
```

```
## Standard Error: 6.760197
```

#part c

```
regmodel <- lm(cd420 ~ treat + age + wtkg + hemo + gender, data = df)
gamma_reg <- coef(regmodel)[2]
var_gamma_reg <- vcovHC(regmodel, type = "HC1")[2, 2]
se_gamma_reg <- sqrt(var_gamma_reg)
cat("Y_hat_reg :", gamma_reg, "\n")
```

```
## Y_hat_reg : 47.7893
```

```
cat("Variance:", var_gamma_reg, "\n")
```

```
## Variance: 45.34205
```

```
cat("Standard Error:", se_gamma_reg, "\n\n")
```

```
## Standard Error: 6.733651
```

#part d

```
intmodel <- lm(cd420 ~ treat * (age + wtkg + hemo + gender), data = df)
gamma_int <- coef(intmodel)["treatCombination"]
var_gamma_int <- vcovHC(intmodel, type = "HC1")["treatCombination", "treatCombination"]
se_gamma_int <- sqrt(var_gamma_int)
cat("Y hat_int (Interacted):", gamma_int, "\n")
```

```
## Y hat_int (Interacted): -64.93356
```

```
cat("Robust Variance:", var_gamma_int, "\n")
```

```
## Robust Variance: 2211.837
```

```
cat("Standard Error:", se_gamma_int, "\n\n")
```

```
## Standard Error: 47.03017
```

#part e TRUE, The difference in means estimator, is the same as the estimate from a regression of the outcome on the treatment and an appropriate estimate of the standard error can be obtained from the OLS summary output. Regression is just comparing the average outcome for treated vs control, which is exactly what the difference in mean does. So, difference in mean will be equal to τ hat. Estimate of the standard error can be obtained by OLS summary output as, as this is a randomized experiment.